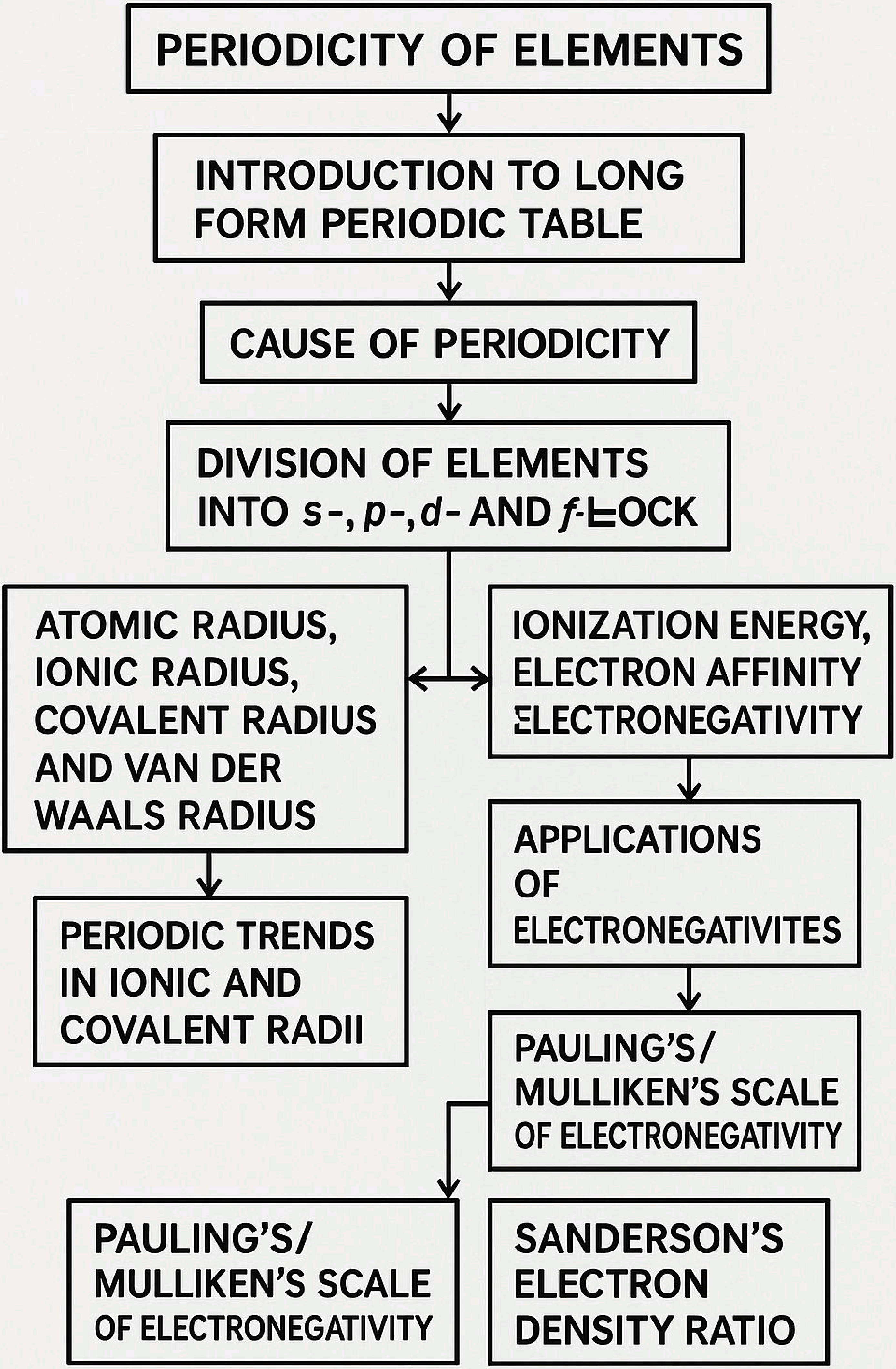

PERIODICITY OF ELEMENTS
INTRODUCTION TO LONG FORM PERIODIC TABLE
CAUSE OF PERIODICITY
DIVISION OF ELEMENTS INTO s-, p-, d- AND f-BLOCK
ATOMIC RADIUS, IONIC RADIUS, COVALENT RADIUS AND VAN DER WAALS RADIUS
IONIZATION ENERGY, ELECTRON AFFINITY ELECTRONEGATIVITY
PERIODIC TRENDS IN IONIC AND COVALENT RADII
APPLICATIONS OF ELECTRONEGATIVITES
PAULING'S/ MULLIKEN'S SCALE OF ELECTRONEGATIVITY
PAULING'S/ MULLIKEN'S SCALE OF ELECTRONEGATIVITY
SANDERSON'S ELECTRON DENSITY RATIO

# PERIODICITY OF ELEMENTS

## LONG FORM PERIODIC TABLE

- Organized by Increasing atomic number
- Shows repeating chemical properties (periodicity)

## CAUSE OF PERIODICITY

- Periodic properties arise due to repeating electronic configurations, especially valence electrons

## CLASSIFICATION OF ELEMENTS

- s-block: Groups 1 & 2
- p-block: Groups 13–16
- *d*-block: Transition elements
- *f*-block: Inner transition elements (lanthanides & actinides)

## IONIZATION ENERGY (IE)

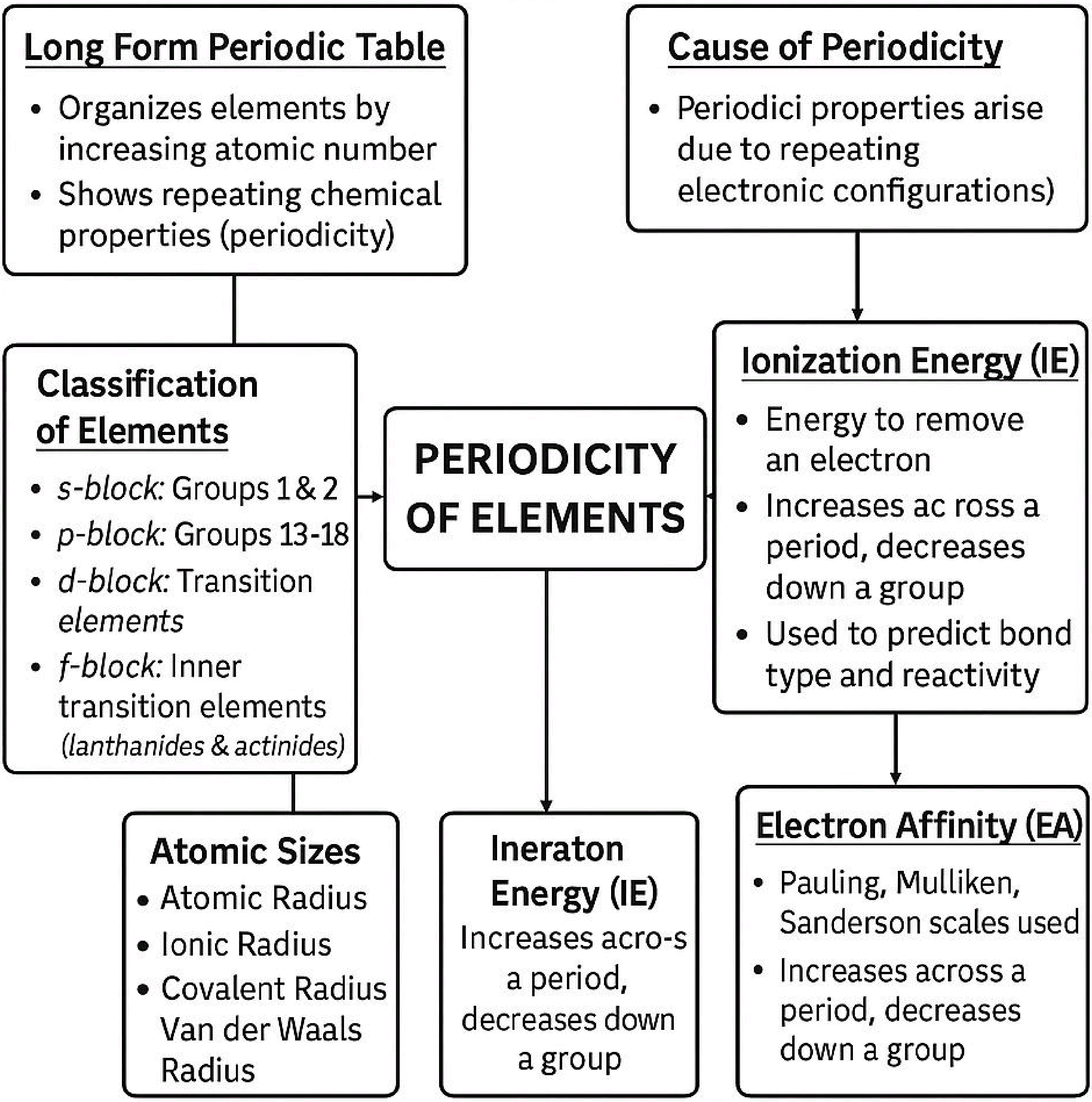
- Energy to remove an electron
- Increases across a period decreases down a group

## FLECTRON AFFINITY (EA)

- Energy change when an atom gains an electron
- More negative across a period, iess negative down a group

## ELECTRONEGATIVITY (EN)

- Tendency to attract electrons in a bond
- Pauling, Mulliken, Sanderson scales used
- EN increases across a period, decreases down a group
- Used to predict


Long Form Periodic Table
• Organizes elements by increasing atomic number
• Shows repeating chemical properties (periodicity)
Cause of Periodicity
• Periodici properties arise due to repeating electronic configurations)
Classification of Elements
• s-block: Groups 1 & 2
• p-block: Groups 13-18
• d-block: Transition elements
• f-block: Inner transition elements (lanthanides & actinides)
PERIODICITY OF ELEMENTS
Ionization Energy (IE)
• Energy to remove an electron
• Increases ac ross a period, decreases down a group
• Used to predict bond type and reactivity
Atomic Sizes
• Atomic Radius
• Ionic Radius
• Covalent Radius Van der Waals Radius
Ineraton Energy (IE)
Increases acro-s a period, decreases down a group
Electron Affinity (EA)
• Pauling, Mulliken, Sanderson scales used
• Increases across a period, decreases down a group

# PERIODICITY OF ELEMENTS

## Long Form Periodic Table

- Arranged by increasing properties across periods, Na, *Mg,* Al in Perid 3 (e.g, Na, Mg. Al in Pelod 3)

## Cause of Periodicity

Due to a repeating electr- configuration (e.g., Na: (Ne]$3s^2$)

## Classification of Elements

- s-block: Groups 1.8.2 (e.g. K)
- *p*-block: Groups 13-18 (e.g. Ci)
- *d*-block: Transition (e.g, Fe)
- *f*-block: Lanthanides (e.g. Ce)

## Atomic Sizes

- Anerg to remove an electron
- Increases across periods, decreases down a group (e.g, Li (520m; Na (496 kJ/mol)

## Ionization Energy (IE)

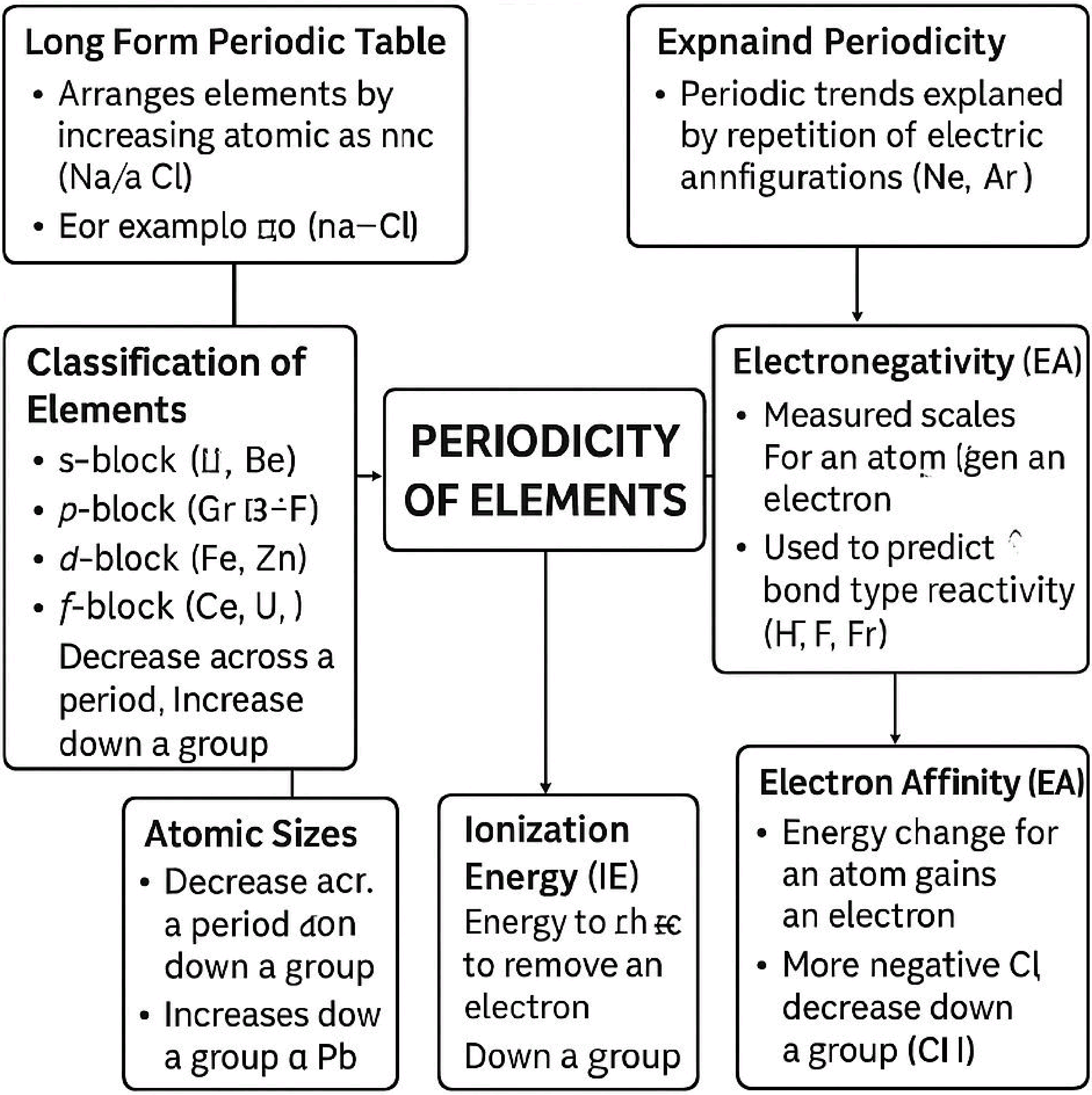
- Energy to remove an electron (e.g, Li 520 kJ/mol) Na (496 kJJ/mol)

## Electronaffinity (EN)

- Ability of an atom to attract electr-
- Increases across periods decreas down a groups (e.g, F: 3,98; Cl: 3.16

## Applications of EN

- Predicting bond type (ionic vs. covalent)
- Determining reactivity (e.g, F highly reactive)
- Determining polarity (e.g. $H_2O$ polar)


Long Form Periodic Table
• Arranges elements by increasing atomic as nnc (Na/a Cl)
• Eor examplo ɑo (na–Cl)
Expnaind Periodicity
• Periodic trends explaned by repetition of electric annfigurations (Ne, Ar)
Classification of Elements
• s-block (Li, Be)
• p-block (Gr B–F)
• d-block (Fe, Zn)
• f-block (Ce, U, )
Decrease across a period, Increase down a group
PERIODICITY OF ELEMENTS
Electronegativity (EA)
• Measured scales For an atom gen an electron
• Used to predict bond type reactivity (H, F, Fr)
Atomic Sizes
• Decrease acr. a period don down a group
• Increases dow a group a Pb
Ionization Energy (IE)
Energy to rh ec to remove an electron
Down a group
Electron Affinity (EA)
• Energy change for an atom gains an electron
• More negative Cl decrease down a group (Cl I)